Final Project Assignment

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CS-332 Data Science

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Introduction

Data Science is the essence of using statistical data and interpreting it for business and research purposes. The data that was given and collected is about the connection between mental health and social media balance. Displaying what participates would rate through every question revolving mental health and social media balance. Some things to take into consideration before going full detail about the data table, understanding the questions and what categories that the table focuses on. Some categories that we will be looking at are, User_ID, Age, Gender, Daily_Screen_Time, Sleep_Quality, Stress_Level, Days_Without_Social_Media, Exercise_Frequency, Social_Media_Platform, Happiness_Index. These categories can help us determine whether an individual has the capacity to balance their mental health and social media. With the given data set, we can determine that the report is based more on young adults to adults. We can assume that with more of the world revolving around technology, the influence of social platforms and the way people can interact with each other throughout the internet can show signs of impact to their livelihoods and affect metal health.

From what I can tell from the data set, the effect of digital platforms and their effect on human beings is getting worse. The effects of being on social media platforms can be reflected on how they rate their sleep quality and exercise. From research on what affects metal health, we can assume that with poor sleep quality, and not feeling physically fit, these can affect the way people mindset what they want. As we can see that the results from the data set, the effects from social media can determine how development of younger people can have more of a negative impact, then positive. Not stating it is only negative, but in some cases, there is a chance that some may find positive from social media. I feel every variable listed in the data set does do a great representation of how social media can affect one's mental health. Stress levels seem to

really be determined by screen time, because as we can see people with less screen time have stated they feel less stressed. This is backed up by the rating they give as feedback to the questions they answered.

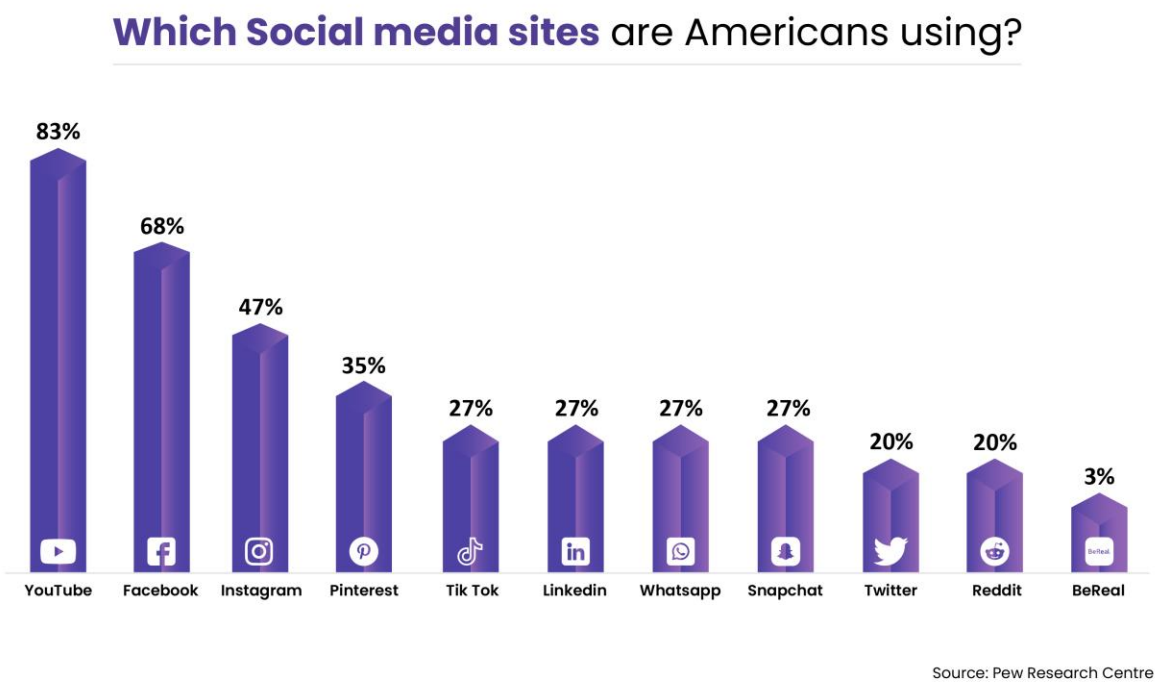


Figure 1: Pew Research Centre on percentages of social media platforms being used by Americans [1]

The connection between mental health and social media can be more impactful than what we perceive. With more of today's society being attracted to electronic devices, the chances of people being on a social media platform are higher. As we will see, social media platforms such as YouTube, TikTok and Facebook will show the difference in age groups and what each age group uses more. With each platform, it would match each age group accordingly based on the when they were created and introduced and what each platform is used for. With YouTube, it was used in a way for any individual to express who and what they want others to see. Such as

gameplay videos for specific games, news, and more. This aligned with younger age groups and some adults. Facebook is the day-old social media platform where the older group uses more. Facebook was known as a new social media platform before the popularity of Instagram, Snapchat, and TikTok. Facebook was a method of interacting with others, sending messages and posting thoughts one may have. In our current era of young generation to young adults the new and trending social media platform is TikTok. As you will see since this is the new popular platform for individuals to promote self, it is well known for individuals creating dance trends and other following and creating these multiple dance skits and now it's one of the most popular platforms for young adults to express themselves.

The connect that can be seen between social media platforms and mental health, is the emotions and feelings a person would feel based on what they are seeing. This could includes from posting content on a platform and getting comments or feedback, this could affect the mental state of the person who posted, based on the comment left any negative comments could hurt the person, cause self doubt and more. This concept could also be implemented to other social media platforms, this is because the when putting your self out in the world, there are people who will judge you based on what they see and believe. However, people who aren’t prepared for the judgement will likely get affected mentally. This is one of the major problems seen in todays world of social media. Judgement on what you believe and what your take is on certian topics of the world.

Questions

1. What is a solution that can help reduce stress?
2. How does the age range affect what they know and believe connects metal health and social media?

3. Do specific social media platforms produce more stress than others?
4. How is happiness scaled on? Is it based on stress, exercise, sleep?
5. What Gender is more affected by social media?
6. Is sleep quality determined by how much screen time they have?
7. Are exercise and stress the only collectable data for mental health?
8. What would be a generalized answer for non-calculated answers?
9. Would having a higher sleep quality affect the amount of stress and exercise they get?
10. Based on specific platforms, is there a difference between social media platforms that produce videos and one that produces image post? How could these two differences affect mental health?

Analysis

About the Data

Link: Mental_Health_and_Social_Media_Balance_Dataset_Kaggle_Gates_dirty.csv

Variables and Types

- User_ID – qualitative
    - A uses id is used to describe the participant with the ability to keep their name anonymous.
- Age – quantitative
    - Age is used to determine the range of which a person has developed and grown.
- Gender – qualitative
    - Is a form of identification that is used to determine if a person is identified as a man or woman.
- Daily_Screen_Time – quantitative
    - This is a measurement of how much an individual spends on a technological screen to visit social media sites.
- Sleep_Quality_1_10 – quantitative
    - This is determined by how they rate how well of sleep quality they are able to get, starting from 1 as least amount of sleep, to 10 to the best amount of sleep.
- Stress_Level_1_10 – quantitative
    - The stress level is labeled between 1 through 10, where 1 is the lowest, and 10 is the highest.
- Days_Without_Social_Media – quantitative

    - This is a measurement of days an individual has gone without the use of social media.
- Exercise_Frequency – quantitative
    - This is described as how much exercise they do, where 0 is no exercise done and 5 as exercises daily.
- Social_Media_Platform – qualitative
    - This category shows the types of social media platforms individual who took the survey responded to as what affected them.
- Happiness_Index_3highest – labeled
    - Names: 1, 2, 3

Gathering and Preparing Labeled Data Via an API

Topics Chosen

YouTube: The reason I chose YouTube as one of the topics is because from my personal experience, I am on YouTube most of the time compared to other social media sites. In theory, it is also one of the most popular video platforms, displaying multiple types of content, targeting a wide variety of audiences.

TikTok: The reason I chose TikTok as another topic is because this is one of the newest platforms for this generation. This is where most of the younger audience would likely be on; therefore, it is a great topic for collecting data on young adults to middle-aged adults. Gathering data on the current generation can be useful for predication on what the future may hold for young adults and their use of social media and the connection to mental health.

Facebook: This topic was chosen for the sole purpose of capturing the audience of middle-aged to even older adults. Since there has been a narrative that most older adults are more on

Facebook than other social media platforms, it is because of how old the platform is. This allows for great data collection, because while the other topics are more targeted towards younger adults, this topic targets the older generation.

Steps to get Data

The method of which I took to get to this point is following guides and researching how to grab data from an Api. First, I started by making a URL with the designated topics that I wanted with my personal Api key. Next, after finding the raw version of data I wanted to collect, I moved to Python coding to help collect the data. This allows for cleaning and organization to collect data, and then the program moves them into csv files for data frames. Then after the collected data is stored in a proper csv file, I use another program to count the recurrence of a word, which allows for displaying how many times a word is used by using a number system to express how many times it occurs in the article. Lastly, with all the collected words that were used within each topic, the next program creates a pdf of images that contain the topic and words that were used within them.

Images

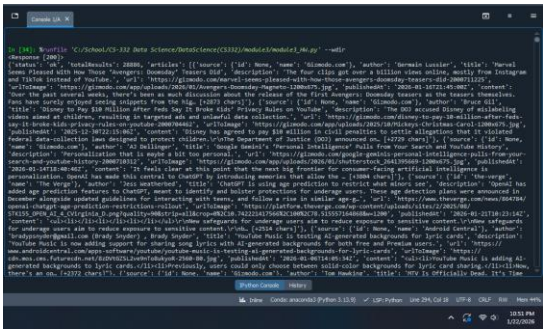


Image 1: Raw Json version of Topics

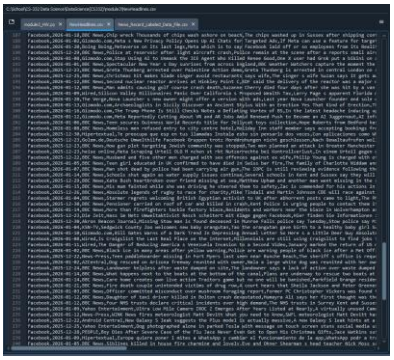


Image 2: CSV of YouTube

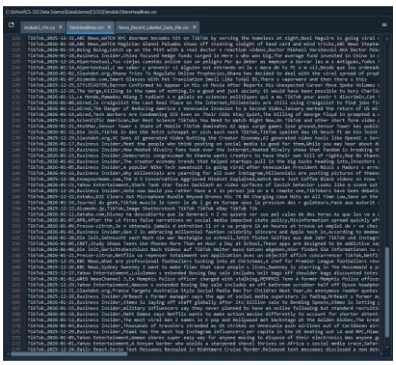


Image 3: CSV of TikTok

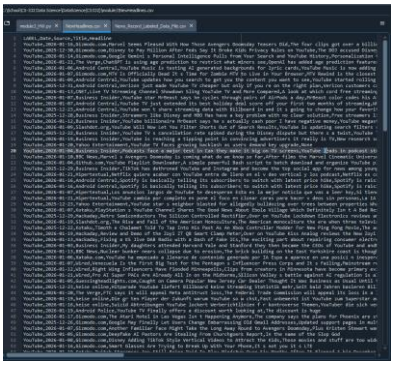


Image 4: CSV of Facebook

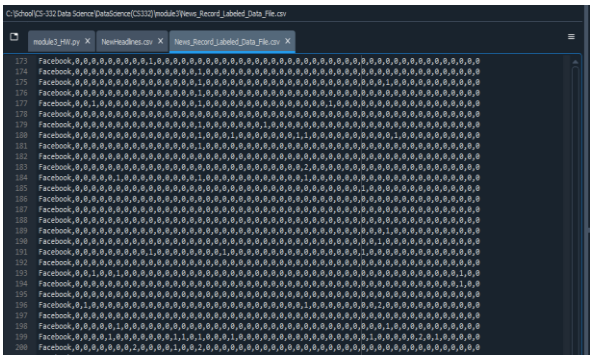


Image 5: CSV of Labeled Data YouTube

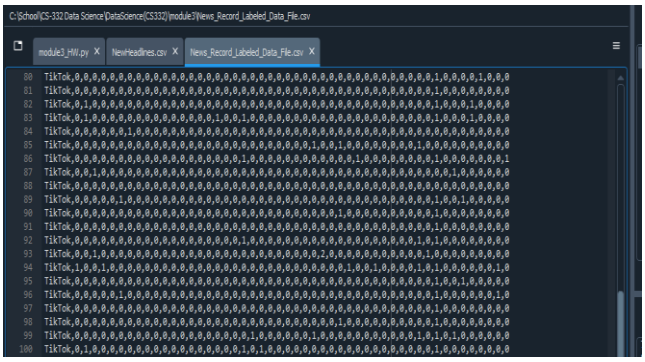


Image 6: CSV of Labeled Data TikTok

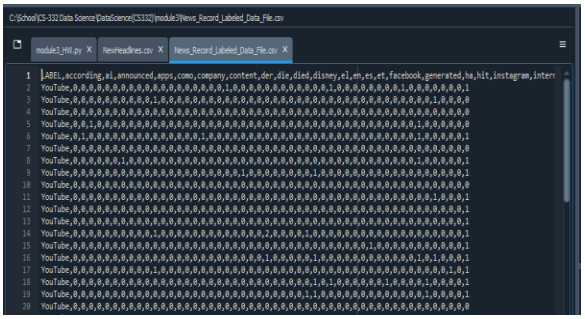


Image 7: CSV of Label Data Facebook

Preparing and Cleaning the Mental Health Dataset

Description: After gathering raw data, there are incorrect or errors in the data. These all need to be looked at carefully and adjusted/fixed to have correct and updated data. With our dataset on mental health levels, I will be fixing each individual column to make sure no errors, missing values, and blanks are represented in the cleaned version of the dataset.

Before Image:

User_ID	Age	Gender	Daily_Screen_Time	Sleep_Quality_1_10	Stress_Level_1_10	Days_Without_Socia	Exercise_Frequency	Social_Media_Platfo	Happiness_Index_1to
U001	44	Male	3.1	7	6	2	5	Facebook	3
U002	30	Other	5.1	7	8	5	3	LinkedIn	3
U003	23	Other	7.4	6	7	1	3	YouTube	1
U004	36	Female	5.7	7	8	1	1	TikTok	2
U005	34	Female	7	4	7	5	1	X (Twitter)	2
U006	38	Male	6.6	5	7	4	3	LinkedIn	2
U007	26	Female	7.8	4	8	2	0	TikTok	2
U008	26	Female	7.4	5	6	1	4	Instagram	2
U009	39	Male	4.7	7	7	6	1	YouTube	3
U010	39	Female	6.6	6	8	0	2	Facebook	2
U011	18	Female	2.8	7	6	2	0	Instagram	2
U012	37	Other	5.4	5	7	3	2	Instagram	3
3456	17	Female	7	7	high	7	1	YouTube	2
U014	39	Female	5.7	5	7	4	0	Facebook	2
U015	45	Male	6.3	7	7	4	3	X (Twitter)	3
U016	17	Female	5.1	7	6	2	5	LinkedIn	3
U017	36	Female	7.5	5	8	4	4	Facebook	2
U018	48	Male	5.4	6	4	3	4	TikTok	3
U019	27	Male	4.7	6	6	0	2	Instagram	3
U020	37	Male	7.1	6	6	5	4	TikTok	3
U021	140	Female	3	8	5	5	4	X (Twitter)	3
U022	42	Other	6.4	7	6	5	3	TikTok	3
U023	43	Male	3.6	8	5	3	1	Facebook	3
U024	31	Female	7.8	5	8	1	2	TikTok	2
U025	30	banana	5	6	6	none	5	X (Twitter)	2
U026	18	Other	5.7	6	7	3	2	X (Twitter)	2
U027	22	Female	6.2	5	8	5	0	X (Twitter)	1
U028	36	Female	2	8	4	5	0	Facebook	3
U029	24	Male	4.3	6	6	5	2	X (Twitter)	3
U030	33	Male	7.4	6	10	3	4	Instagram	2
U031	1	Male	7.1	5	8	5	2	X (Twitter)	3
U032	40	Male	6	6	7	5	4	TikTok	2
	29	Female	3.1	8	4	2	2	LinkedIn	3
U034	24	Female	6.6	6	7	4	0	TikTok	2
U035	41	Other	6.8	8	7	2	2	LinkedIn	3

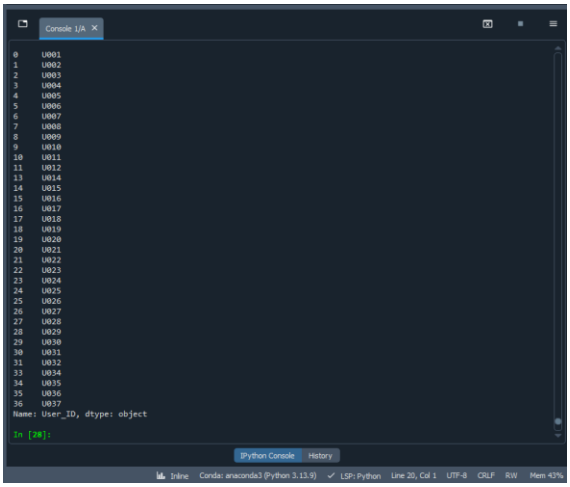
To first start visually, we can see that the first columns have some errors in the id. Invalid blank ids can become a problem with data, because it could affect authenticity, not ensuring that the data that was collected from a set individual will line up with the topic at hand. To first start the cleaning process, I will use pandas to first read the csv file and print out the first 15-25 lines of data and with that I can narrow down, print out the first column only and then remove any id that are invalid.

Before:

```
Console 1/A

0     U001
1     U002
2     U003
3     U004
4     U005
5     U006
6     U007
7     U008
8     U009
9     U010
10    U011
11    U012
12    3456
13    U014
14    U015
15    U016
16    U017
17    U018
18    U019
19    U020
20    U021
21    U022
22    U023
23    U024
24    U025
25    U026
26    U027
27    U028
28    U029
29    U030
30    U031
31    U032
32     NaN
33    U034
34    U035
Name: User_ID, dtype: object

In [10]:

IPython Console   History

Inline   Conda: anaconda3 (Python 3.13.9)   LSP: Python   Line 19, Col 24   UTF-8   CRLF   RW   Mem 47%
```

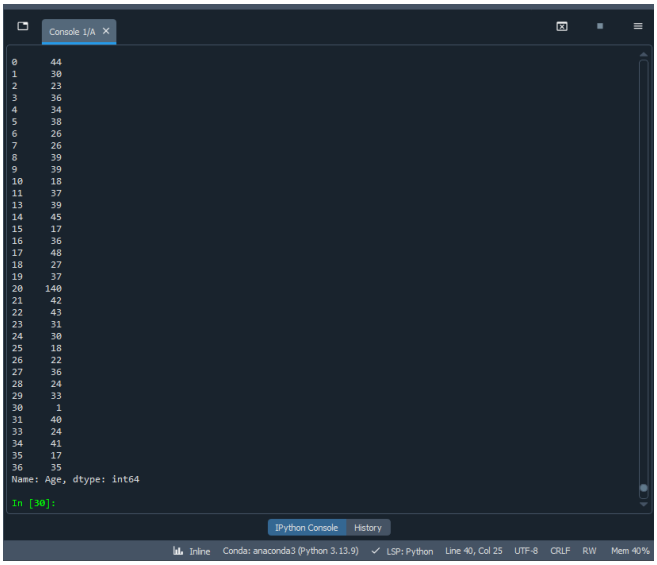
As we can see from the before image, there are some missing and invalid user ids that need to be removed. To do so, I will be utilizing the renaming function to change any invalid ids to NaN, then use the dropna() function to remove them from the set. This will remove any NA/NAN values, in this image you will notice the number 12 and 32 are missing, and that is because it was removed, for invalid id.

After Image:



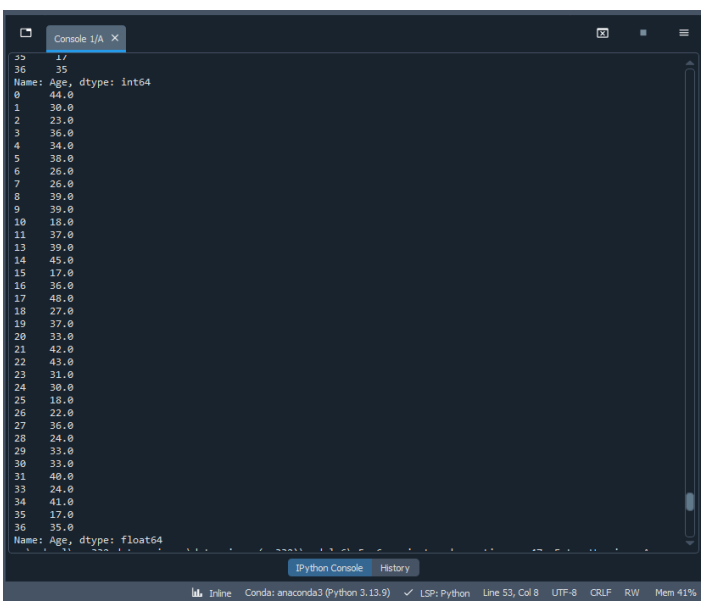
The next column we can now look at here is the “Age” column, where it displays the age of everyone. To get clear and understandable data, I believe the age limit of this data was between 16-50, this is because I believe most of metal health is occurring during this age group as you younger than 16, the toll of mental health is not as affective on an individual, due to the fact that they may not be able to fully comprehend what is metal health. As for individuals who are older than 50, they have faced many challenges of metal health and have found ways to decrease the effect of metal health on them.

Before:



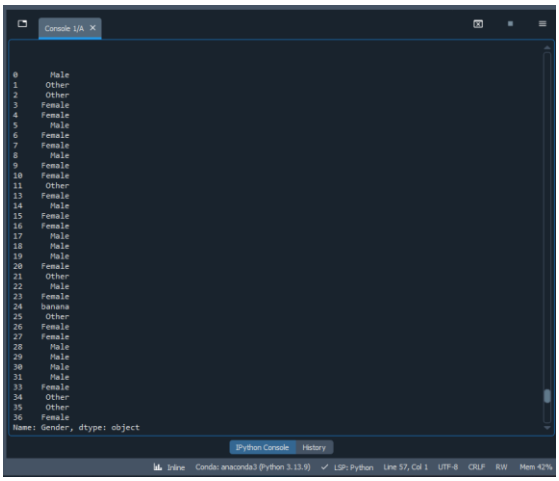
So, moving forward in this column, my goal is to remove the values lower than 16 and higher than 50 and replace them with the averages of the whole column. This is because although their age might have been a typo, looking at their other data, it seems to me that it was just a common error, and not a huge mistake that should remove the whole row. Here we will also change this data type to an integer, to clean up the data to be aligned with clarity.

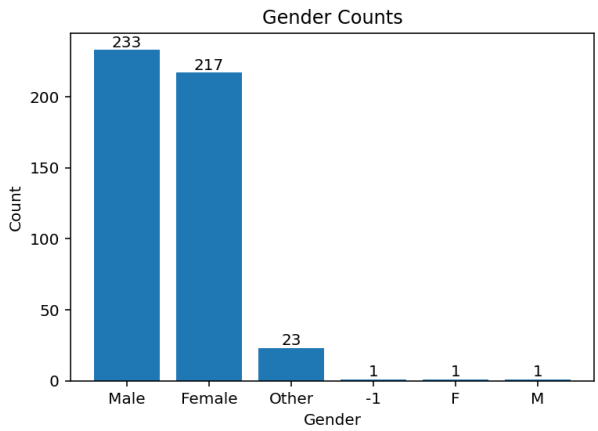
After:



For this next column, I will be working with Gender. There are 3 options for the genders, Female, Male, and Others. Any other value that is inputted into the dataset is considered invalid or a wrong answer given.

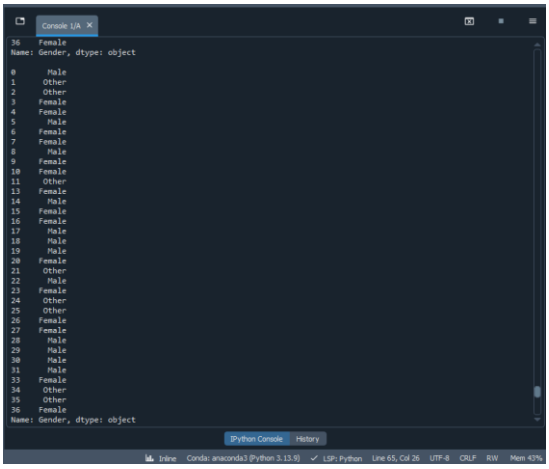
Before Image:

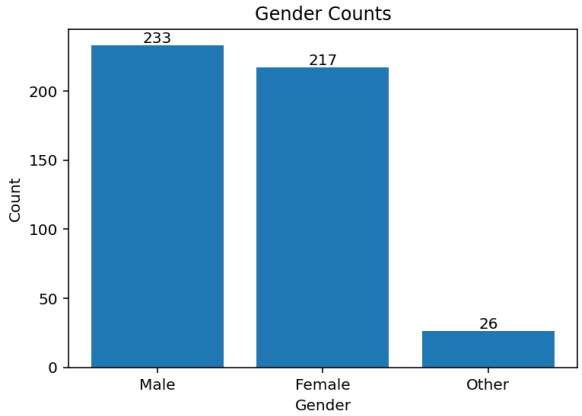




The goal here is to remove any invalid answer and replace them with others, since that is considered a way of not assuming what gender they are, but giving a proper answer.

After Image:

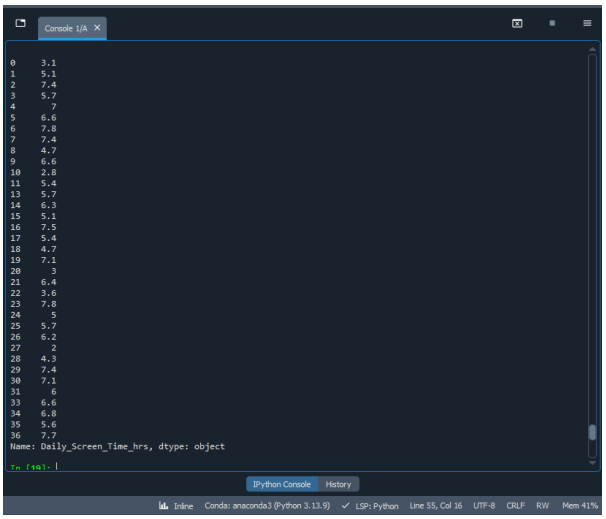


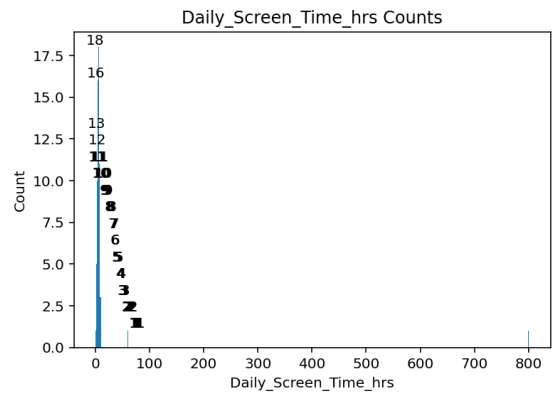


In column Daily_Screen_Tme_hrs, there are some error values and blanks values. The objective of this column is to clean up the column by making the data all unison. Make sure it stays as a float, and all projected values are projected as a float. What we will be doing here is

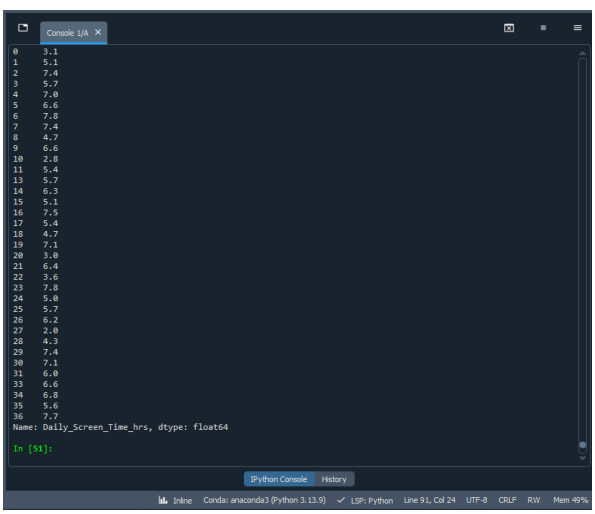
making sure the values are shown as a float and then remove any rows that have invalid values that are not within the parameters of the requirements.

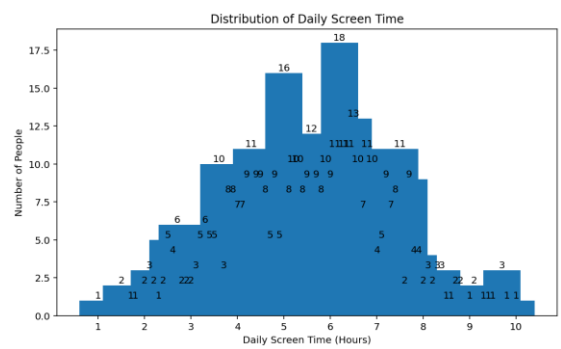
Before Image:





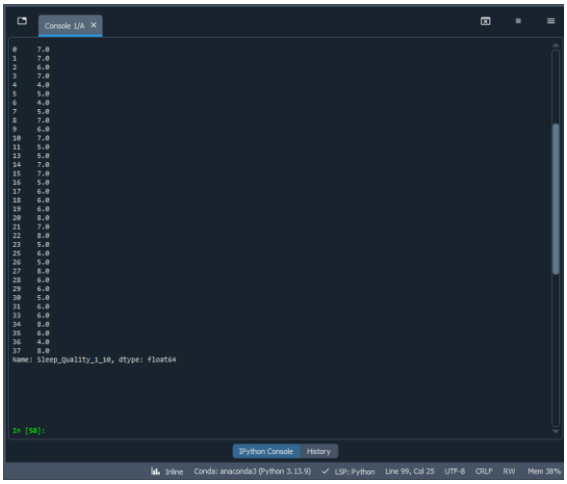
After Image:

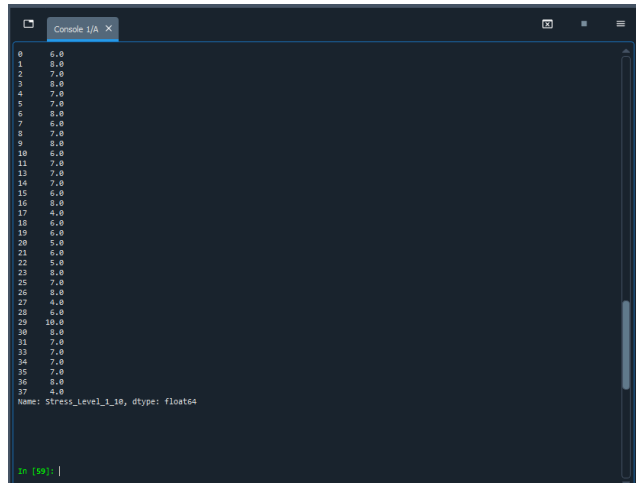


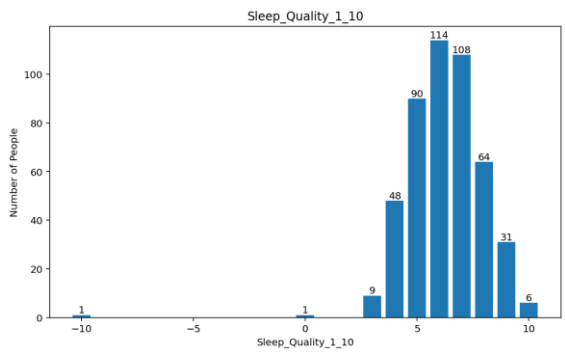


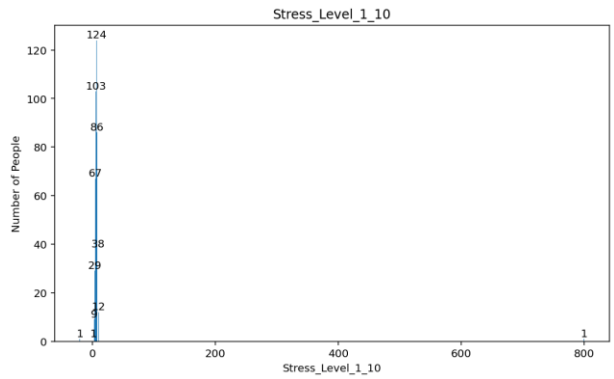
With the next 2 columns on Stress Quality and Level, the cleaning is going to be similar cleaning wise. The objective of this column is to clean up the invalid data that are either blank or incorrect values. Knowing that it uses a parameter of 1-10, we can remove all values that do not meet the required parameter.

Before Image: 1. Stress Quality 2. Stress Level



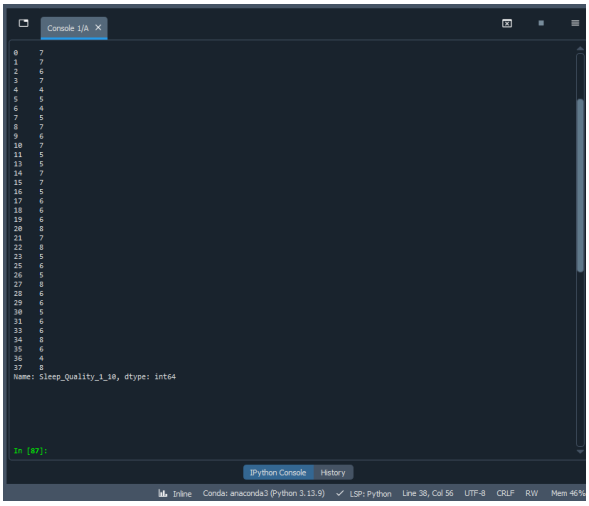


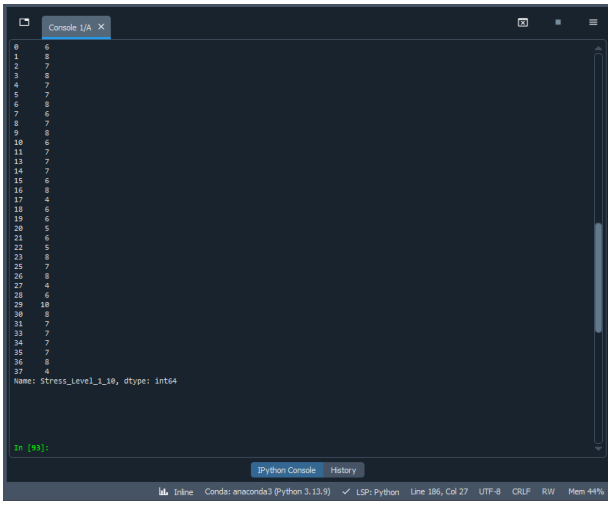


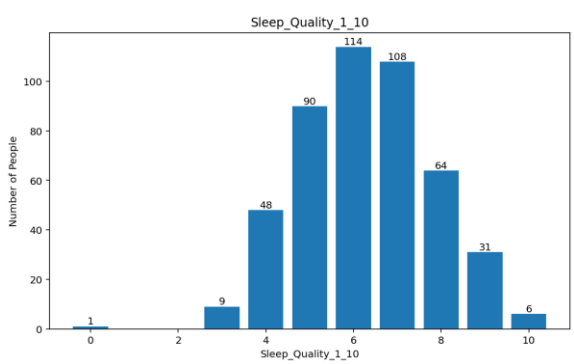


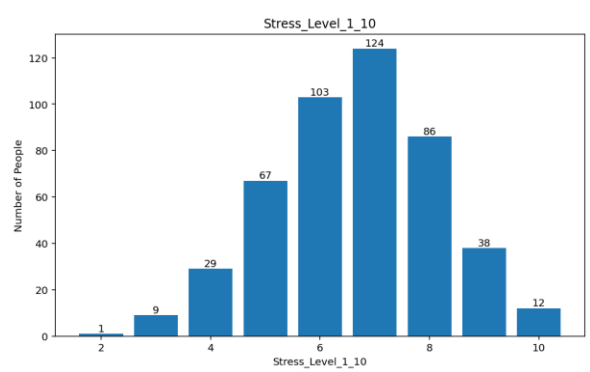
As you can see from the first 35 prints, the data is in float form. The goal here I believe is to make the data type to an integer with no decimal values. I am also going to double check the data on making sure that the data is within the parameters of our limits 1-10. Next, I will remove any invalid values and print the results of the change.

After Image: 1. Stress Quality 2. Stress Level



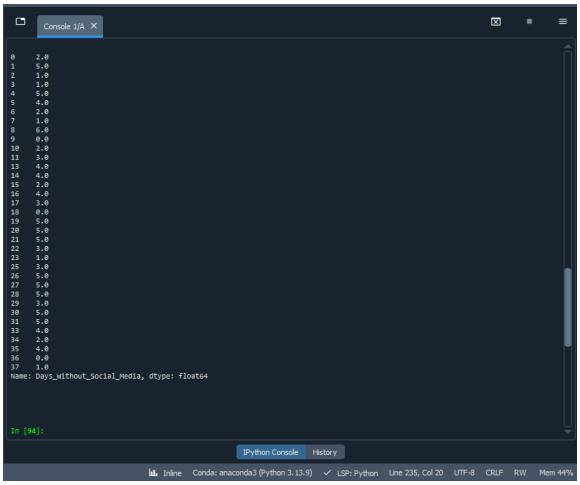


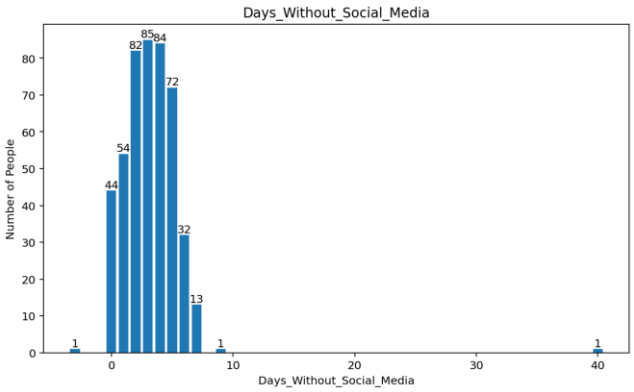




The next column we will be looking at here is what each individual goes without social media. In this column, the focus is on the number of days an individual has chosen to take off not being on social media. So based on this, we know our parameters for valid values are between 0-7 since there are only 7 days a week.

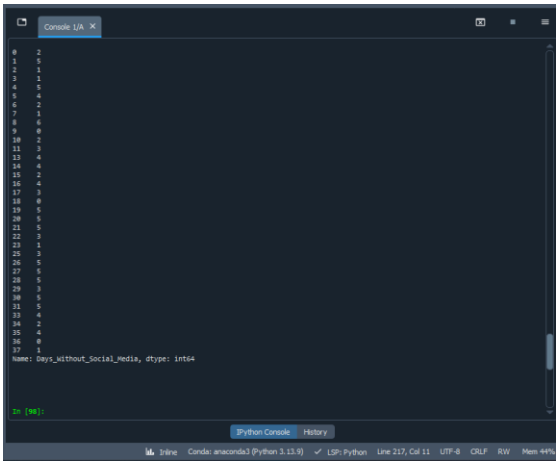
Before Image:

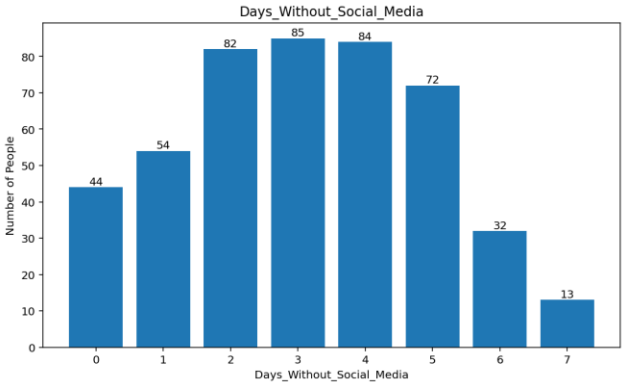




What I did here was to make sure the values are integers and display only values that are within the 0–7-day range. To do this, I will utilize the same functions as before, by replacing and dropping invalid values.

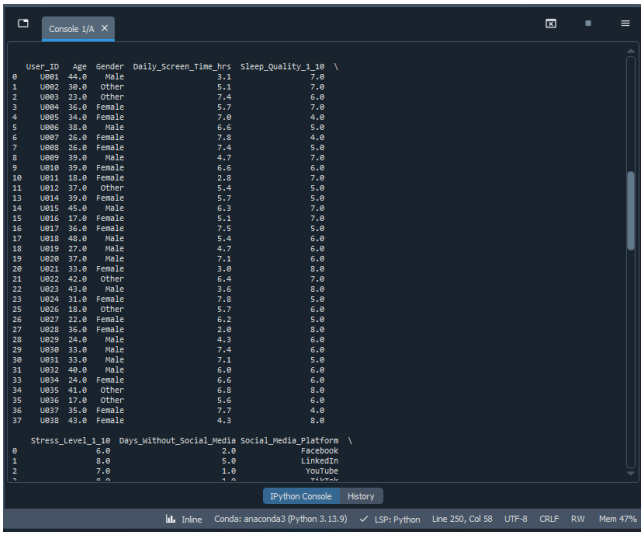
After Image:





With this next column Exercise_Frequency, I chose to remove this from the data set, because I see there is a little connection to mental health, but I wouldn't think of it to have as big of an impact as the other columns. This column only shows the level of excersie level they day but doesn't show the difference from when their mental health was higher or lower, so the information just seems unnecessary. I will use the drop function to remove the column.

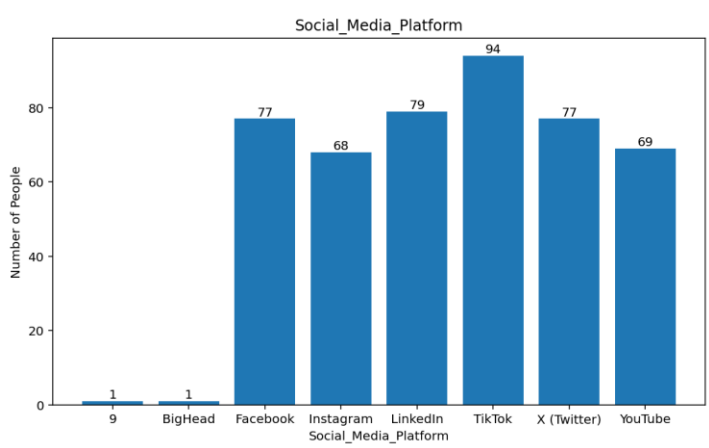
After Image:



```
Console 1/A

   User_ID   Age  Gender  Daily_Screen_Time_hrs  Sleep_Quality_1_10  \
0     U001  44.0    Male                    3.1                 7.0
1     U002  30.0   Other                    5.1                 7.0
2     U003  23.0   Other                    7.4                 6.0
3     U004  36.0  Female                    5.7                 7.0
4     U005  34.0  Female                    7.0                 4.0
5     U006  38.0    Male                    6.6                 5.0
6     U007  26.0  Female                    7.8                 4.0
7     U008  26.0  Female                    7.4                 5.0
8     U009  39.0    Male                    4.7                 7.0
9     U010  39.0  Female                    6.6                 6.0
10    U011  18.0  Female                    2.8                 7.0
11    U012  37.0   Other                    5.4                 5.0
13    U014  39.0  Female                    5.7                 5.0
14    U015  45.0    Male                    6.3                 7.0
15    U016  17.0  Female                    5.1                 7.0
16    U017  36.0  Female                    7.5                 5.0
17    U018  48.0    Male                    5.4                 6.0
18    U019  27.0    Male                    4.7                 6.0
19    U020  37.0    Male                    7.1                 6.0
20    U021  33.0  Female                    3.0                 8.0
21    U022  42.0   Other                    6.4                 7.0
22    U023  43.0    Male                    3.6                 8.0
23    U024  31.0  Female                    7.8                 5.0
25    U026  18.0   Other                    5.7                 6.0
26    U027  22.0  Female                    6.2                 5.0
27    U028  36.0  Female                    2.0                 8.0
28    U029  24.0    Male                    4.3                 6.0
29    U030  33.0    Male                    7.4                 6.0
30    U031  33.0    Male                    7.1                 5.0
31    U032  40.0    Male                    6.0                 6.0
33    U034  24.0  Female                    6.6                 6.0
34    U035  41.0   Other                    6.8                 8.0
35    U036  17.0   Other                    5.6                 6.0
36    U037  35.0  Female                    7.7                 4.0
37    U038  43.0  Female                    4.3                 8.0

    Stress_Level_1_10  Days_without_Social_Media Social_Media_Platform  \
0                 6.0                        2.0               Facebook
1                 8.0                        5.0              LinkedIn
2                 7.0                        1.0               YouTube

IPython Console   History

Inline   Conda: anaconda3 (Python 3.13.9)   LSP: Python   Line 250, Col 58   UTF-8   CRLF   RW   Mem 47%
```

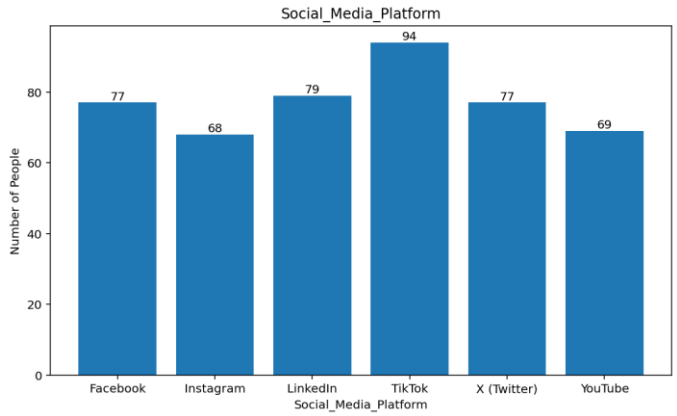
This next column is a list of social media platforms. The column I believe is important, because based on what age group and the connection between the social media platforms we can see that there is a difference between which social media platform each age group uses. For this column, I am going to use a Bar chart to gather what is being collected. To determine if they are picking proper social media platforms.

Before Image:



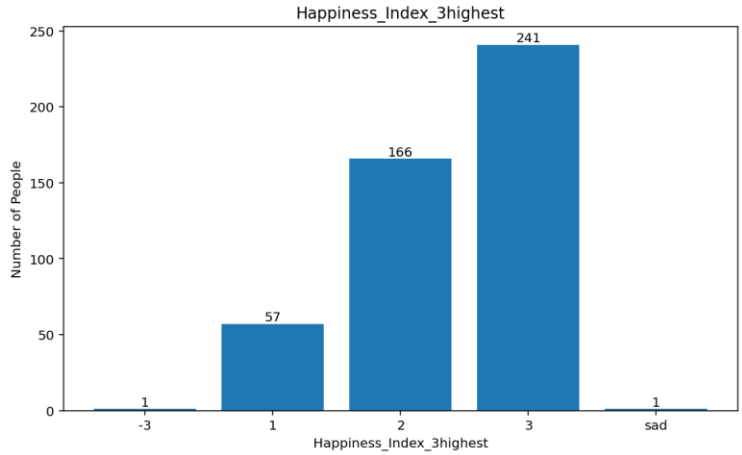
As we can see, there are 2 non-social media platforms. To get rid of these, I will create a list that contains the two non-social media platforms and replace them with NaN or NA values so that I can remove them using the dropna() function.

After Image:

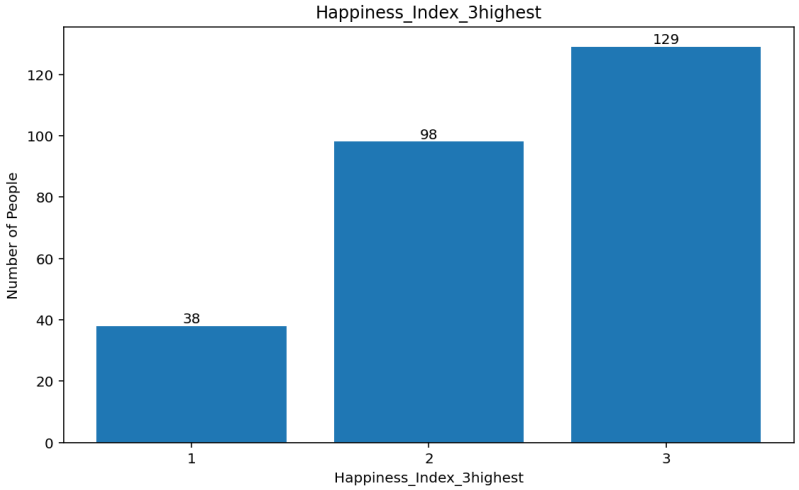


The last column I have here is the Happiness Index, which is the scale on which a person expresses their happiness from 1-3. So, this already gives us the parameters that we want to build on and develop. Looking at the data, we can see that there are some values that are negative and non-integer representation of numbers. What I am going to do is first get a visual data chart on the column to see how much of the data is only 1-3 and what I need to remove.

Before Image:

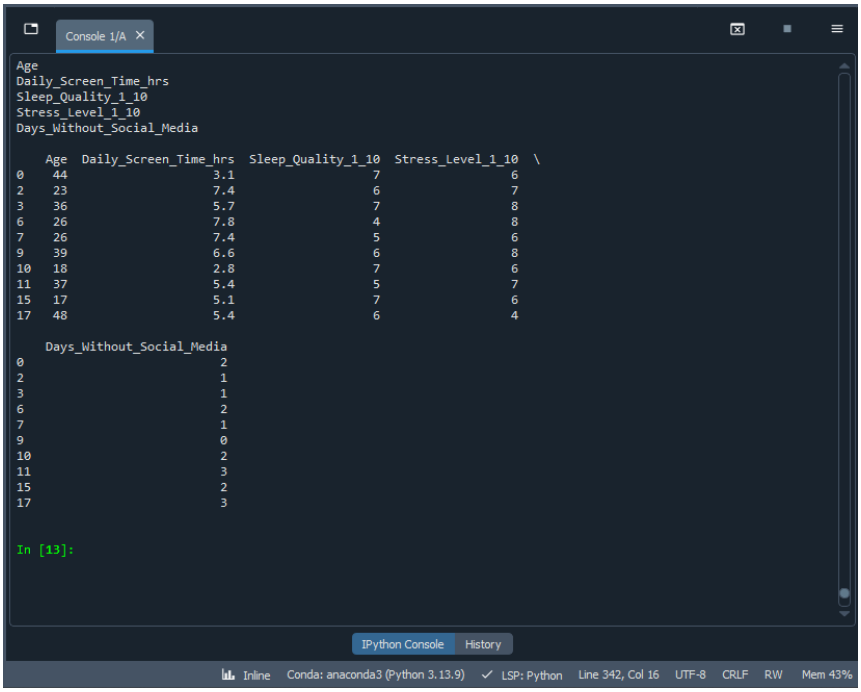


After Image:



The last step here is to demonstrate our understanding of unlabeled data to create a new data frame that consists of only unlabeled and quantitative data. To do this, we need to determine what is quantitative and unlabeled. I know that this data frame label is the happiness index, so the columns to remove are going to be UserID, Gender, Social_Media_Platform, and Happiness_Index_3highest. As you can see in the image below, only the columns that are unlabeled and quantitative are left.

After Image:



```
Console 1/A

Age
Daily_Screen_Time_hrs
Sleep_Quality_1_10
Stress_Level_1_10
Days_Without_Social_Media

    Age  Daily_Screen_Time_hrs  Sleep_Quality_1_10  Stress_Level_1_10  \
0    44                    3.1                   7                  6
2    23                    7.4                   6                  7
3    36                    5.7                   7                  8
6    26                    7.8                   4                  8
7    26                    7.4                   5                  6
9    39                    6.6                   6                  8
10   18                    2.8                   7                  6
11   37                    5.4                   5                  7
15   17                    5.1                   7                  6
17   48                    5.4                   6                  4

    Days_Without_Social_Media
0                           2
2                           1
3                           1
6                           2
7                           1
9                           0
10                          2
11                          3
15                          2
17                          3


In [13]:

IPython Console   History

Inline   Conda: anaconda3 (Python 3.13.9)   LSP: Python   Line 342, Col 16   UTF-8   CRLF   RW   Mem 43%
```

Unsupervised Learning with KMeans Clustering

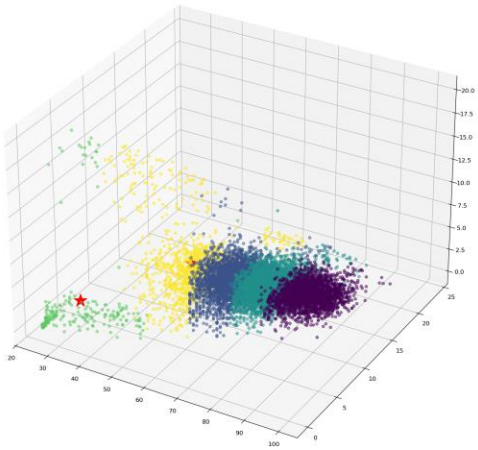
Mental Health Data Set

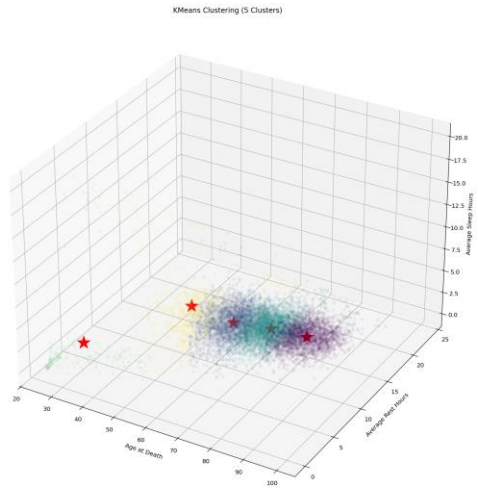
Description: In this section of the project report, I will be using KMeans to cluster up the data to find out what similarities and differences the data has. First, we will need to define the amount of cluster to use, and to find this, we can start with a base number of 5, because since we have 5 total columns that are quantitative.

First we will need to have our program read the csv file and then we will need to drop the first column “unamed: 0”. This is useless and it was added when creating it as a new csv file. After reading and dropping the csv file, next we will need to make sure we import KMeans from sklearn, and then we can start compiling the sklearn and plotting the visual images. We will be using the scatter plot graph from matplotlib to visualize the clustering effects.

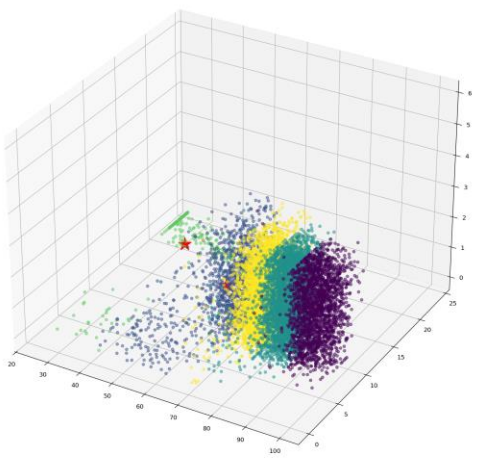
Images:

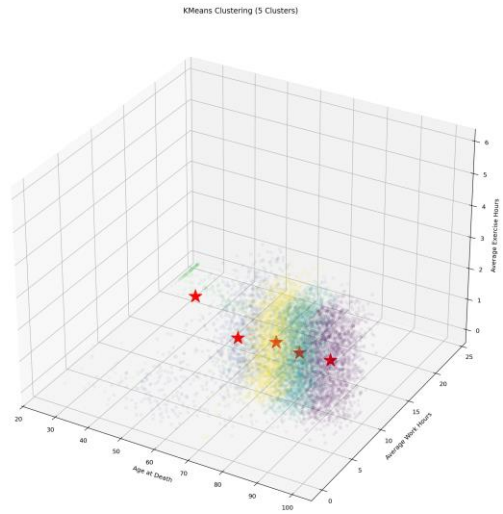
(Age, Rest, Sleep)





(Age, Work, Exercise)





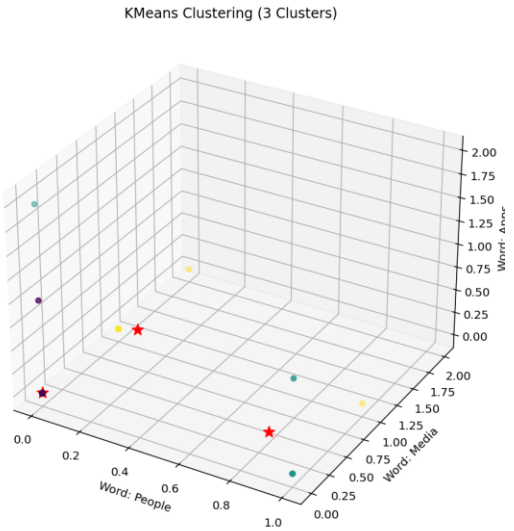
Results Review: From looking at the graphs, we can see that the clustering is very interesting in our center points where the n_clusters show that they are in the middle of each clustering. What we can take from this graph is that the clustering follows a similar pipeline, which is indicated by how the layers are set. With 5 clusters, it looks relatively correct on how it is all spread out evenly. I have found that as the older the ages are, the more likely they are to have higher mental health issues. This is perceived, because we can see from the graphs that initially started with

younger people not having to deal as much with mental health at higher levels, compared to older adults.

News API Data Set

Description: In this data set, we will be clustering the data that contains words that are related to mental health that are connected to social media websites. This data was collected from a list of articles that contain words that are related to mental health. We will be using KMeans clustering on this data set to see what different social media platforms have in common with one another. To first start KMeans clustering, we need to make sure the data is cleaned to where no labeled and qualitative data are present. To do this, we will use pandas to read the csv file and display the data frame and remove the label name "LABEL", which stands for the social media platform that is connected to each world that is in set article. I am going to cluster 3 columns and these will be people, media, and apps. This will hopefully show us the connection between the 3 topics of choice. Since we are using 3 topics, I will try using 3 clusters to see what the results are and find connecting dots.

Image:



As we can see from the clustering, we can notice that there is not much, however this shows us that the connections between the clusters are not as close, because not all the articles found with the word, didn't contain the word. What we can take from this is that the article collected did not have much similarity with one another, in sharing similar words. In contrast, compared to our first use of KMeans, there is a lot of similarity to what it connects to.

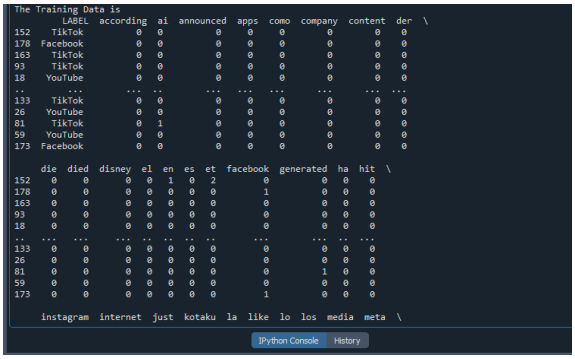
Supervised Learning with Decision Trees

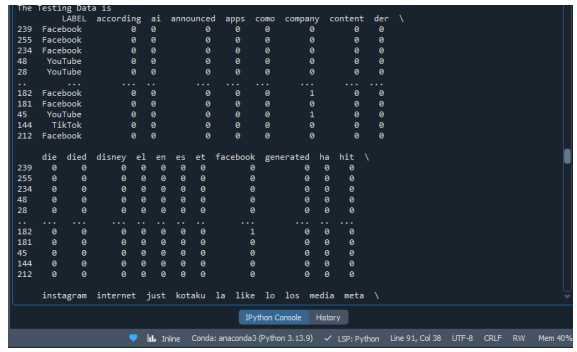
News API Dataset:

With this dataset, it focuses on gathering data on specific websites that speak on social media platforms and the connections to mental health. The data will contain data on what words are found within the website, and what social media platform it talks about. We will first need to read the csv file containing the data on the News API and then apply the split function of training and testing data. This will then allow us to work on building our decision tree; that will show us the similarity of the two-split data.

To first start the process of creating our decision tree with this dataset, we need to create our train and test datasets, and then with that set our labels and we will check the datasets to make sure that the actions we have done are complete. After creating our two sets, we will use them in the decision of trees to create a visualization of the datasets. What we will see in the images of creating the datasets is that it will contain the label and data of true and false on if an image was seen in the news article.

Images:



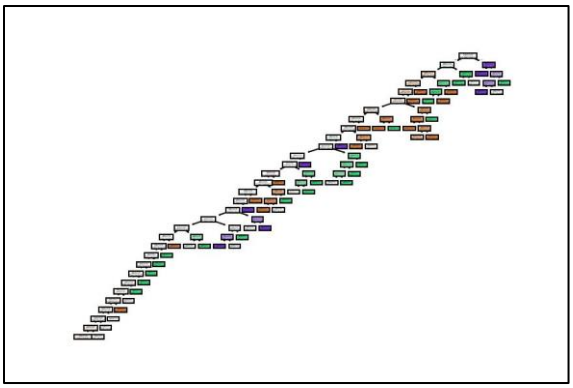


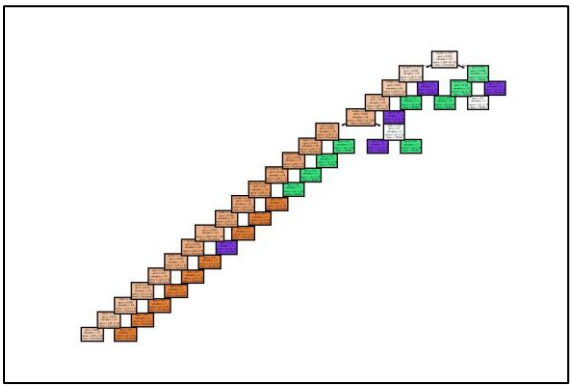
Next we will store the label in a separate section that will be used for the decision tree plotting. With that, once the labels are set in their own section, we will delete the label column in the data frame to allow for clean use of decision tree plotting. This will include using the drop method in pandas to delete the columns of our choice, which in this case is the “LABEL” column.

Label Removal Image:

After we have set out our variables, we can now apply the decision tree plotting to get our visualized figure. Using the function plt to plot our figure, we will plot the decision tree and save it as a jpg file. In the plot, we will need to set our variable names and what is being collected to create the image. For this instance, I will be using the train and test datasets and use the decision tree classifier to call the classes names. Making sure the decision is full and filled in and making sure the image fits all in 1 image, with all the data.

Decision Tree Image:





From the Image that we can see some similarities and differences. This is already an indication that the data that was collected have something in common, and this statement is based on the observation of how the tree focuses on moving in 1 main direction and split to the other side once every section. Showing that in every new article it contained 1 or 2 words within them.

Mental Health Dataset:

To begin digesting the dataset, we will need to first divide the dataset into only quantitative and labeled data. With this, we will be able to start Supervised Learning, using decision trees and train and testing data. After using pandas to read the csv file, and then removing the non-quantitative data, I will now need to introduce a new label based on the label feedback on the dataset. The new label that is introduced is the visual explanation of what is categorizing the happiness levels, using labels names such as unhappy, okay, and happy. Then after initializing the label, we will then remove the happiness index column that uses label names of 1, 2, and 3.

After the data is now prepped and ready, we can now go on to the next phase of applying the supervised learning methods. To start, we want to have our adjusted dataset be called and make sure the dataset is correct and in order. Then the next step is to create our test and train data. This will allow us to do manual testing and prediction testing from the function to predict the data. This will allow us to compare the results and the accuracy of the prediction, to see any

information that would be useful such as similarities, even distribution, and more. After dividing the dataset into training and testing data, we will next set up the labels of training and testing data into its own separate section. This will allow us to remove the label from the dataset, while keeping the labels separated and stored in a separate entity.

Images:

```
Console 1/A

The Training Data is
      Age  Daily_Screen_Time_hrs  Days_Without_Social_Media Happieness_Label
121   32                    3.6                          3            Happy
205   19                    2.7                          2            Happy
225   33                    6.9                          0             Okay
128   20                    5.1                          2            Happy
186   20                    5.0                          3            Happy
..   ...                    ...                        ...              ...
25    29                    4.0                          3            Happy
260   41                    7.7                          2             Okay
239   28                    4.4                          3             Okay
238   18                    4.2                          2            Happy
29    41                    5.6                          2            Happy

[197 rows x 4 columns]
The Testing Data is
      Age  Daily_Screen_Time_hrs  Days_Without_Social_Media Happieness_Label
145   44                    6.5                          2             Okay
112   40                    6.4                          1             Okay
33    16                    6.9                          3          Unhappy
257   23                    3.3                          2            Happy
258   39                    3.0                          1            Happy
..   ...                    ...                        ...              ...
44    29                    4.0                          1            Happy
70    39                    5.6                          2            Happy
150   35                    6.1                          3            Happy
88    48                    8.5                          3             Okay
61    37                    4.3                          3            Happy
```

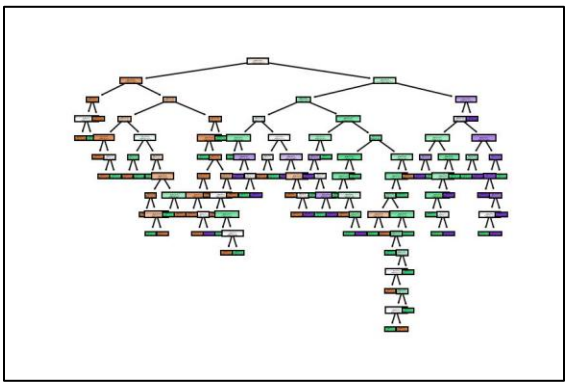
```
[66 rows x 4 columns]
121    Happy
205    Happy
225     Okay
128    Happy
186    Happy
        ...
25     Happy
260     Okay
239     Okay
238    Happy
29     Happy
Name: Happieness_Label, Length: 197, dtype: category
Categories (3, object): ['Unhappy' < 'Okay' < 'Happy']
This is Train Lablels
      Age  Daily_Screen_Time_hrs  Days_Without_Social_Media
121   32                    3.6                          3
205   19                    2.7                          2
225   33                    6.9                          0
128   20                    5.1                          2
186   20                    5.0                          3
..   ...                    ...                        ...
25    29                    4.0                          3
260   41                    7.7                          2
239   28                    4.4                          3
238   18                    4.2                          2
29    41                    5.6                          2
```

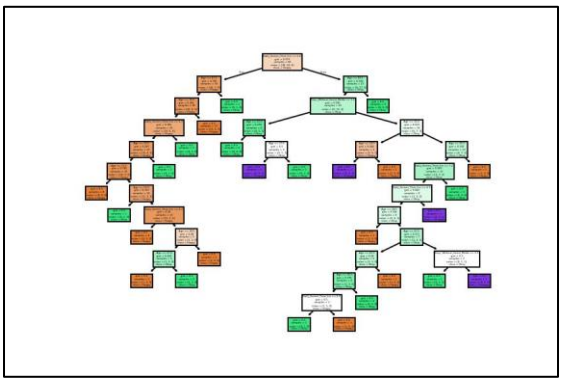
```
[197 rows x 3 columns]
145      Okay
112      Okay
33     Unhappy
257     Happy
258     Happy
        ...
44      Happy
70      Happy
150     Happy
88       Okay
61      Happy
Name: Happieness_Label, Length: 66, dtype: category
Categories (3, object): ['Unhappy' < 'Okay' < 'Happy']
This is Test Lablels
      Age  Daily_Screen_Time_hrs  Days_Without_Social_Media
145   44                    6.5                          2
112   40                    6.4                          1
33    16                    6.9                          3
257   23                    3.3                          2
258   39                    3.0                          1
..   ...                    ...                        ...
44    29                    4.0                          1
70    39                    5.6                          2
150   35                    6.1                          3
88    48                    8.5                          3
61    37                    4.3                          3

[66 rows x 3 columns]
```

The next step is to apply the decsion tree plotting to get a visualization of what is happening with our current data. This will allow us to see and compare the accuracy of the predication with the use of the confusion matrix. As we can see from the image of the decsion

tree, we will notice that the accuracy is relatively correct, because based on the answer that was found the predictions are closely accurate to what the label matches.

Images:





My conclusion here is that the prediction from the findings, we can say that the distribution was not the best, however the flow of the decision tree is relatively the same. The decision tree shows us the similarities of the prediction to what we train them to be, and we can see that all the data falls within the same concept of where the origin is and what connects to what.

Conclusion:

With all the information we have just walked through and looked at, the ability to connect mental health and social media should be a little easier to understand and make sense. To overview the paper, we have talked about what data is/means, methods/ways to collect data using an API method, cleaning and prepping the dataset, and both unsupervised and supervised learning. Through every process we learn increasingly what the connection between social media platforms and mental health. While most of the time speaking about mental health and social media, it seems to be more of the negative aspects of it. We can also see the use of social media to help promote awareness about mental health. Data can be represented as information that is collected that is useful to help prove and provide information on topics.

While working through the data that was collected and collecting the data, making sure that the data is relevant to the topic of choice is also crucial to reliable information. As we have done, you will see we used filtering functions to filter out the data that is our target and then store it as collective data of importance. In this project, I have gathered data from a news Api, focusing on gathering data from only the social media of our target. In my case, it was correct to explain what social media platforms are widely used from different age groups. I chose to filter the news from Api to only gather data that was from YouTube, Facebook, and TikTok. A refresher on the reason behind it is because from the mental health table we use, some of the major used social media platforms ranging from all age groups; these were the best targeted data.

Our process of cleaning and preparing data using pandas was an interesting and fun way to learn methods of cleaning and prepping. Using pandas to read csv files and displaying a data frame to see the errors and invalid values. This is crucial to make sure the data is validated and contains only values that fall under the category's valid inputs. As it is important to validate the values/inputs, making sure to get a better understanding of what we are working with is also true. Learning the difference between variables that are labels, qualitative, and quantitative. This information would be useful to apply to correcting the input values data type. Making sure that each and every values data type matches the column that we are cleaning. After fixing the mistakes and errors, it will be useful for applying them onto unsupervised and supervised learning.

With this project, we have worked with two different learning methods to explain the datasets that we have cleaned. With these learning methods, it will allow for a better understanding of what is happening within the dataset with the use of visualizations. Seeing the visualization will help brainstorm ideas and meanings of what the dataset could have in common

and what they don't. With the use of unsupervised learning methods, we showed the method of clustering and that allowed us to see what groups of individuals have in common and what they differentiate. With clustering we are able to see what groups each individual falls within and what the average cluster point is to genialize the similarity. The other method we use and look at is the supervised method to create decision trees that shows the connections from the node and the roots that connect to them. After seeing how we collect data and findings of the connectivity between social media and mental health. Something we can do every day to help increase positive mental health; is to take our time off electronic devices and spend time with people you care about the most. The use of social media could also be used for better than worse, by providing positive mentality to improving mental health.

Image: Mental Health Awareness

Reference

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